

Michael Kelso Glass

Senior Application Developer / UX-Focused Systems Builder

Modernizing Internal Tools | UX-Focused Systems | Messy Operational Problems | Portland-bound

Profile

I'm a tool-agnostic application developer with a strong front-end and UX focus. I like messy problems: aging systems, confusing workflows, slow queries, unfamiliar platforms, and projects where the path forward is not obvious yet.

My career has moved through government, consulting, publishing, internal systems, technical writing, teaching, and application architecture. The pattern has stayed pretty consistent: learn the system, understand the people using it, find the friction, and make the tool clearer, faster, and easier to trust.

I enjoy front-end work because I like seeing ideas become usable. But I do not want to be boxed into one layer of the stack. I'm at my best when UX, architecture, troubleshooting, and practical problem-solving overlap.

What I Bring

- Modernize aging internal applications without losing sight of the people who depend on them
 - Design front-end experiences that make complex workflows easier to understand and use
 - Learn unfamiliar platforms quickly and become productive without waiting for perfect documentation
 - Untangle performance bottlenecks, brittle logic, and "nobody knows why this happens" problems
 - Translate operational complexity into clear, maintainable application workflows
 - Create reusable front-end standards, patterns, and conventions for future development
 - Help teammates solve problems by explaining the reasoning, not just handing over the answer
 - Use AI-assisted workflows to understand unfamiliar systems faster, reduce grunt work, and speed up documentation, coding, and idea development
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Selected Work

Contra Costa County EHSD

Information Systems Programmer / Analyst IV

November 2012 – Present

Administrative Department, Application Development

Highest-level role in the ISPA classification, focused on application development, front-end modernization, internal systems, workflow tools, and long-term support for county operational applications. Started as a Robert Half contractor in March 2012, continued as a county temporary employee in November 2012, and became a full-time county employee in October 2013.

Workload Distribution Tool (WDT)

Took over a critical workload distribution project after the original vendor was removed midway through development. Led the remaining development and support for a .NET / SQL Server application used by roughly 1,000 eligibility and social work staff.

The system distributed work based on staff availability, skill sets, bilingual capabilities, vacation schedules, and other operational constraints. It also integrated with CalWIN data, including read-only Oracle-backed data sources that were painfully slow when I inherited them.

Reworked the data access and query approach enough to reduce some of the worst-running calls from minutes, and in some cases closer to an hour, down to a few seconds. WDT stayed in production for several years with very few true application defects. The system did what internal tools are supposed to do: it solved the problem and kept working.

STARS.NEXT / Knowledge Management Modernization

Contributing front-end UX work for STARS.NEXT, a new knowledge management system currently in concept/prototype development. The long-term goal is to turn STARS into an AI-driven application that helps county employees find information, documents, and operational knowledge more easily.

My focus is the front-end experience: modernizing the interface, improving usability, and helping define standards we can reuse on future projects. The work is also serving as a test case for newer county development patterns, including Azure-hosted .NET MVC applications with stronger CSS and JavaScript support.

CFS Wraparound System

Architected and developed an internal application used to track services and payments connected to foster care and child welfare programs. The direct user group was small, but the system supported staff managing information tied to thousands of children receiving county services.

Handled the full development lifecycle, including requirements, database design, application architecture, front-end design, development, testing, and support.

Trust Fund Management System

Designed and developed an internal financial tracking system for managing trust fund allocations for children under county care.

The county holds the funds in a shared Wells Fargo account, while the application tracks each child's individual entitlement, balance, interest, and related activity. The system gave staff a clearer, more reliable way to manage funds that were not separated at the bank-account level.

Earlier Career Highlights

PHP Publishing / Technical Authoring

Wiley Publishing

Beginning 2003, with later revision work

Lead author and developer for *Beginning PHP, MySQL, and Apache Web Development* and *Beginning PHP5, MySQL, and Apache Web Development*. Came to PHP through a real business need: replacing Lotus Notes based internal tools with web applications that were easier to deploy and maintain. Wiley later asked me to lead a team of authors on a PHP development book after I became active helping other developers solve problems in the PHP community.

Selected Previous Roles

Robert Half Technology / Consulting — Software Architect / Consultant

March 2008 – October 2012, various consulting roles

Designed and developed web applications for public-sector, nonprofit, and private clients using ASP.NET, C#, VB.NET, PHP, SQL Server, JavaScript, HTML/CSS, and related web technologies.

Corp of Enabled Veterans — Lead Software Architect

April 2010 – August 2012

Led architecture and development for web-based and Windows applications, including requirements, design, implementation, testing, documentation, maintenance, and change management.

Disabled American Veterans — Software Engineer II

July 2008 – April 2010

Developed and maintained internal fundraising applications using ASP.NET/VB and Oracle-backed systems.

Fifth Third Bank — Senior Data Architect / Release Manager

July 2004 – June 2007

Managed builds, releases, deployment coordination, ClearCase/ClearQuest release packages, implementation planning, and business continuity/disaster recovery planning for core banking applications.

Reach Publishing — Director, Software Development

January 2001 – December 2002

Managed and developed internal sales, HR, CRM, and order-entry systems. Led the transition away from Lotus Notes-based applications toward PHP/MySQL web applications.

Whittman-Hart / MarchFirst — Senior Consultant

January 1999 – July 2001

Worked on enterprise and startup web applications, including PlanetFeedback.com using Vignette StoryServer/Tcl, and Lotus/Domino-based applications for corporate clients. Frequently assigned to unfamiliar technologies based on ability to learn quickly and contribute fast.

Synergistics — Lead Developer / Knowledge Management Architect

August 1997 – December 1998

Led development and architecture for a Knowledge Management product during the early KM wave in the Lotus/Domino ecosystem, including requirements, technical design, development, documentation, and QA methodology.

Miller Heiman — IT Manager

April 1995 – July 1997

Supported network and internal systems, administered company technology, and began the transition into software development by creating Lotus Notes-based versions of Strategic Selling and Conceptual Selling tools.

Technologies & Platforms

Primary: ASP.NET / C#, HTML, CSS, JavaScript, SQL Server, MVC architecture

Current / Recent: ServiceNow Portal, Azure-oriented .NET MVC applications, front-end standards, UX modernization, AI-assisted workflows

Additional Background: PHP, MySQL, Oracle-backed systems, Lotus Notes/Domino, Vignette StoryServer/Tcl, ClearCase/ClearQuest, Visual Basic, MS Access modernization

Focus Areas: Internal tools, legacy modernization, workflow systems, front-end usability, application architecture, performance troubleshooting, technical writing

Writing, Teaching & Knowledge Sharing

I've always learned best by building things, breaking them down, and explaining them to other people. That pattern shows up in a few different places: published technical author, longtime pool instructor, informal problem-solver for coworkers, and now frequent user of AI-assisted workflows for writing, coding, documentation, and idea development.

I don't see teaching as separate from development. It's part of how I understand systems. If I can explain the problem clearly, I'm usually a lot closer to solving it.

Military Service

U.S. Navy — Machinist's Mate Second Class

August 1985 – October 1993

Served aboard USS Carl Vinson (CVN-70) and USS Sacramento (AOE-1), including service during Operation Desert Shield and Operation Desert Storm.

While aboard USS Sacramento, built a Quattro Pro-based tool to automate fuel tank calculations used during underway replenishment planning. The tool reduced turnaround time for hog-and-sag calculations from roughly six hours to as little as 15 minutes, helping leadership make faster decisions about fuel distribution.

Worked in high-responsibility operational environments built around reliability, maintenance discipline, teamwork, and practical problem-solving.